

SUMMARY

Machine Learning / Research Engineer with 8+ years of experience bridging research and production in robotics and autonomy. Focused on foundation-model-based representation learning, multimodal alignment, and agentic systems for long-tail scenario discovery and safety evaluation. Proven track record designing purpose-driven metrics, scalable evaluation pipelines, and real-world deployments that ensure systems meet their true safety objectives.

EDUCATION

- **Georgia Institute of Technology** Atlanta, GA
Ph.D. in Robotics and M.S. in Computer Science; GPA: 4.00/4.00 Aug 2017 - Dec 2023
- **Texas A&M University** College Station, TX
B.S. in Physics and B.S. in Electrical Engineering; GPA: 4.00/4.00 Aug 2012 - May 2017

RESEARCH EXPERIENCE

- **Robot Perception and Learning Lab** Atlanta, GA
Graduate Research Assistant - Advisor: Dr. Zsolt Kira Aug 2018 - Dec 2023
 - [ICRA'20] Captured, curated, and released multi-agent dataset for deep-learned visual collaboration.
 - [CVPR'20] Co-founded research domain of [collaborative perception](#). Leveraged attention for inter-agent communication.
 - [IROS'21] Enhanced semantic segmentation with patch attention and spatial registration between moving robots.
 - [ICRA'23] Mined safety-critical scenarios for self-driving cars. Reduced collisions with distributed BEV motion forecasts.
 - [ICRA'24] Improved crop yield measurements with large-scale, distributed image search.
- **Mitsubishi Electric Research Labs** Boston, MA
Research Intern, Multi-Robot SLAM - Advisor: Dr. Milutin Pajovic Summer 2019
 - [Patent] Extended particle filter paradigm to multi-robot SLAM via efficient communication of particles.

INDUSTRY EXPERIENCE

- **Zoox (self-driving for Amazon)** Foster City, CA
Software Engineer Jan 2024 - Present
 - Co-developed a foundation-model-based **scene embedding and retrieval** stack for long-tail scenario mining in autonomous robotaxi operations, supporting safety-critical collision avoidance validation.
 - Tech lead for **True Positive (TP) collision avoidance** evaluation, ensuring collision avoidance intervenes when required; defined purpose-driven safety metrics beyond aggregate performance statistics and reported progress week-over-week.
 - Coordinated across multiple teams to curate **new synthetic safety-critical scenarios** spanning new driving domains and weather patterns, improving coverage of rare but consequential failure modes.
 - Built **scalable simulation and evaluation pipelines** enabling routine large-scale testing for about **20 developers**, improving iteration speed and consistency of safety validation workflows.
 - Conducted industry research to enhance embedding-based lookup for scene similarity; post-trained **autoregressive** and **masked autoencoder (MAE)** foundation models on autonomy logs to learn robust scene representations.
 - Developed highly optimized, parallelized **JAX** evaluation functions for model introspection of **reinforcement-learning-learned trajectories**, enabling efficient analysis at scale.
 - Built **agentic scenario triage** tooling leveraging LLMs and VLMs to provide rapid triage metrics; implemented **text-to-SQL** and agentic crawlers over heterogeneous internal data sources to discover novel root-cause failure modes.
 - Developed dataset sourcing and **simulation dataset compression** methods; validated via week-over-week longitudinal studies, reducing simulation cost from **\$100k/month to \$10k/month** while preserving metric sensitivity.
- **Bowery Farming** Sunnyvale, CA
Senior Software Engineer Feb 2022 - Dec 2023
 - Co-developed and deployed vision-based quality control systems across **3 commercial farms (439k sq ft total)**, reducing labor and material waste at production scale.
 - Designed novel algorithms for large-scale image correspondence and retrieval enabling farm-wide analytics across **1000+ grow positions**, each with **independent robotic lighting and irrigation controllers**.
 - Built data and evaluation pipelines to analyze behavior across heterogeneous robotic subsystems, enabling detection of anomalies and regressions across farms; presented related work at **ICRA'24**.
 - Owned end-to-end ML lifecycle including dataset design, model training, deployment, and monitoring in real-world environments.
- **Traptic, Inc. (acquired by Bowery Farming)** Sunnyvale, CA
Software Engineer Mar 2020 - Feb 2022
 - Gave sight to robotic harvesting with 2D and 3D vision algorithms. Curated datasets, trained, and deployed. [\[video\]](#)

INTERNSHIPS

- **R-Dex Systems** Atlanta, GA
Intern, Warehouse SLAM Summer 2018
 - Improved SLAM accuracy in GPS-denied environments using domain-specific computer vision.
- **Southwest Research Institute** San Antonio, TX
Intern, Unmanned Ground Vehicles Summer 2016, Summer 2017
 - Developed localization hardware and software for off-road self-driving.
- **Lockheed Martin Aeronautics** Fort Worth, TX
Intern, F-16 System Test and Integration Summer 2015

CONSULTING EXPERIENCE

- **Phood Solutions**: Improved machine learning for vision-based food cataloging. Handling edge cases with VLM fallback.
- **Tennibot**: Conducted R&D of SLAM algorithms for tennis court localization.

SELECTED PUBLICATIONS

- [1] **N. Glaser**, R. Ravi, Z. Kira. N-QR: Natural Quick Response Codes for Multi-Robot Instance Correspondence, *International Conference on Robotics and Automation (ICRA)*, 2024
- [2] **N. Glaser**, Z. Kira. We Need to Talk: Identifying and Overcoming Communication-Critical Scenarios for Self-Driving, *International Conference on Robotics and Automation Workshops (ICRA Workshops)*, 2023
- [3] **N. Glaser**, Z. Kira. Communication-Critical Planning via Multi-Agent Trajectory Exchange, *International Conference on Robotics and Automation (ICRA)*, 2023
- [4] **N. Glaser**, Y.-C. Liu, J. Tian, Z. Kira. Overcoming Obstructions via Bandwidth-Limited Multi-Agent Spatial Handshaking, *International Conference on Intelligent Robots and Systems (IROS)*, 2021
- [5] **N. Glaser**, Y.-C. Liu, J. Tian, Z. Kira. Enhancing Multi-Robot Perception via Learned Data Association, *International Conference on Robotics and Automation Workshops (ICRA Workshops)*, 2020
- [6] Y.-C. Liu, *J. Tian, ***N. Glaser**, Z. Kira. When2com: Multi-Agent Collaborative Perception via Communication Graph Grouping, *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020
- [7] Y.-C. Liu, J. Tian, C.-Y. Ma, **N. Glaser**, C.-W. Kuo, Z. Kira. Who2com: Collaborative Perception via Learnable Handshake communication, *International Conference on Robotics and Automation (ICRA)*, 2020
- [8] J. Tian, W. Chung, **N. Glaser**, Y.-C. Liu, Z. Kira. UNO: Uncertainty-aware Noisy-Or Multimodal Fusion for Unanticipated Input Degradation, *International Conference on Robotics and Automation (ICRA)*, 2020
- [9] J. Tian, Y.-C. Liu, **N. Glaser**, Y.-C. Hsu, Z. Kira. Posterior Re-calibration for Imbalanced Datasets, *Conference on Neural Information Processing Systems (NeurIPS)*, 2020

SKILLS SUMMARY

- **Languages**: Python, C++, C, Java, SQL, Matlab, Scala, Pyspark
- **Tools**: Pytorch, Tensorflow, Jax, Databricks, Linux, ROS, Docker, Git, Unity, AWS, Flask

COMPETITIONS

- **AlphaPilot** (sponsored by Lockheed Martin and Drone Racing League) Atlanta, GA
Perception Team Lead Spring 2019 - Fall 2020
 - **High Speed Indoor Drone Racing**: Helped implement gate detectors (Tensorflow, YOLO), robust visual-inertial odometry (ROVIO), and factor graph pose optimization (GTSAM) algorithms for high-speed autonomous drone racing.
 - **Outcome**: Top 9 (out of 420 teams); qualified to compete during physical race trials held across the country
- **Mohamed Bin Zayed International Robotics Challenge** Atlanta, GA
Perception Team Lead Fall 2018 - Spring 2020
 - **Multi-Agent Perception, Localization, and Coordination**: Implemented detectors (Tensorflow) for various challenge objects (drones, balloons, and bricks) and integrated robust visual-inertial odometry to complement intermittent GPS.
 - **Outcome**: Submission awarded \$100k to compete in Abu Dhabi, UAE
- **Texas Instruments Innovation Challenge** Atlanta, GA
Team Lead Fall 2015 - Spring 2016
 - **Powered, Programmable Elbow Orthosis**: Processed biopotential signals to drive exoskeleton motors per a user's intended movements. Devised efficient signal processing algorithms for implementation on a low-power microcontroller.
 - **Outcome**: 2nd place; \$7500 award

PROJECTS

- [llm.markets](#): Stock market analysis tool for quickly backtesting your “crackpot” market theories (e.g. hurricane $\uparrow$ NYSE:HD).
- [Circuit Component Detector](#): Analyzed performance of deep object detector trained with varying regimens of synthetic and real data. Stitched internet-scraped images of circuit components and added realism with a generative adversarial network.
- [Online Center of Mass Estimation](#): Implemented robust control and online learning for 17-DoF inverted pendulum robot.
- [Railgun](#): Displayed power of invisible electromagnetic forces by launching projectile with velocity of a small handgun.
- [Robotic Arm](#): Created anatomically-accurate, optically-controlled robotic hand.

HONORS AND AWARDS

- Second place in [ICRA'23 CoPerception](#) workshop; oral presentation
- Spotlight talk at [ICRA'20 EDAT](#) workshop
- First place [senior design project](#) in Texas A&M ECE; second place (\$7500 award) in TI Innovation Challenge
- Received [top academic award](#) at Texas A&M, only one awarded per department per year
- [National Merit Finalist](#); received full presidential scholarship to Texas A&M